HU Collection
TMUA Mock Set B Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	C	C	C	B	E	B	C	A	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	A	D	E	C	E	C	A	C	E	A

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **C**.

Look for a quantity a move does not change. Since $\dfrac{1}{ab/(a+b)} = \dfrac{1}{a} + \dfrac{1}{b}$, the sum of the reciprocals of the numbers on the board is invariant. Initially it is $1 + 2 + \cdots + 20 = 210$, so the final number has reciprocal 210, whatever order the moves are made in. The answer is $\frac{1}{210}$. Option F is the answer for someone who tries a couple of orders, gets messy fractions, and gives up.

Question 2. Answer **C**.

Work with remainders of powers of 2. Modulo 3 the powers go $2, 1, 2, 1, \ldots$, so $2^{\text{even}} \equiv 1$ and I holds. Modulo 5 the cycle is $2, 4, 3, 1$ of length 4; $2026 \equiv 2 \pmod 4$, so $N \equiv 4$ and $N + 1 \equiv 0$: II holds. Modulo 7 the cycle is $2, 4, 1$ of length 3; $2026 \equiv 1 \pmod 3$, so $N \equiv 2$ and $N - 1 \equiv 1 \neq 0$: III fails. The trap in III is assuming that because $2^3 \equiv 1$ and 2026 is even, something nice happens; the exponent has to be read modulo the cycle length 3, not 2.

Question 3. Answer **C**.

A number with strictly increasing digits cannot contain 0 (it would have to be the leading digit), and it is determined completely by the set of its digits, which is any non-empty subset of $\{1, \ldots, 9\}$ of size at most 3. So the count is $\binom{9}{1} + \binom{9}{2} + \binom{9}{3} = 9 + 36 + 84 = 129$. Option A counts only the three-digit numbers; option F is all subsets of the nine digits, size unrestricted.

Question 4. Answer **C**.

Which statement is the false one? It cannot be $a < c$: if $a < c$ were false then $c \leq a$, but the other two statements would then be true, giving $a < b < c \leq a$, impossible. So $a < c$ is always among the two true statements, and must be true. Either of the other two can be the false one: $(a, b, c) = (1, 3, 2)$ makes $b < c$ false, and $(2, 1, 3)$ makes $a < b$ false. Those two examples also dispose of D and E.

Question 5. Answer **B**.

Add up every entry in the grid in two ways. Summing the rows gives a sum of n odd numbers, which is odd exactly when n is odd; summing the columns gives a sum of n even numbers, which is always even. So n odd is impossible. For n even, put 1 in every cell of the first column and 0 everywhere else: every row sums to 1 and the columns sum to n (even) and 0. The answer is even n only. Counting one quantity two ways is the whole method.

Question 6. Answer **E**.

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Let $F(c) = \int_0^c f$, so $F(0) = F(2) = 0$. For II, F is differentiable with $F' = f$, and a function that starts and ends at 0 must have zero derivative somewhere in between (if F is not identically zero it has an interior maximum or minimum, where $F' = 0$; if it is identically zero then $f \equiv 0$). So II holds. For III, let $g(c) = \int_c^{c+1} f$ for $0 \le c \le 1$. Then $g(0) + g(1) = \int_0^1 f + \int_1^2 f = 0$, so $g(0)$ and $g(1)$ are of opposite sign or both zero, and since g is continuous it vanishes somewhere on $[0, 1]$. So III holds. I fails: $f(x) = 1 - x$ has $\int_0^2 f = 0$ but $F(c) = c - \frac{1}{2}c^2 = \frac{1}{2}c(2 - c) > 0$ for $0 < c < 2$. The answer is II and III. The idea in III, adding the two half-length integrals to force a sign change, is worth remembering; it is the reason there is always a unit-length interval on which f has zero average.

Question 7. Answer **B**.

$10^{10} = 2^{10}5^{10}$, and factorials have far more factors of 2 than of 5, so only the fives matter. The number of factors of 5 in $n!$ is $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \cdots$. For $n = 44$ this is $8 + 1 = 9$; for $n = 45$ it is $9 + 1 = 10$. So $n = 45$. Option C is the answer to "ten multiples of 5", forgetting that 25 contributes twice.

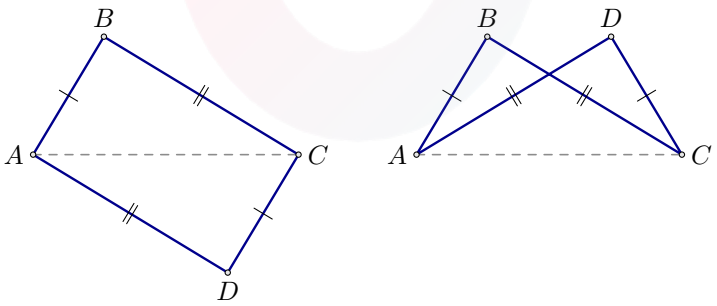
Question 8. Answer **C**.

Two integers have sum or difference divisible by 12 exactly when their remainders on division by 12 are equal or add to 12. Group the remainders into the classes $\{0\}$, $\{6\}$, $\{1, 11\}$, $\{2, 10\}$, $\{3, 9\}$, $\{4, 8\}$, $\{5, 7\}$: seven classes, and two numbers from the same class always work. So 8 integers force a pair. Seven can avoid it: the remainders $0, 1, 2, 3, 4, 5, 6$ are all distinct and no two of them sum to 12. The answer is 8. Option B is the count of classes, one short of the pigeonhole answer.

Question 9. Answer **A**.

A square leaves remainder 0, 1 or 4 on division by 8. Adding three such remainders can give $0, 1, 2, 3, 4, 5, 6$ but never 7 (the largest available combinations are $4 + 4 + 4 \equiv 4$, $4 + 4 + 1 \equiv 1$, $4 + 1 + 1 = 6$, $1 + 1 + 1 = 3$). Since $2023 = 8 \cdot 252 + 7$, there are no solutions at all. No searching is needed, which is the point: the options B to E are the counts one gets after finding one solution and multiplying by sign and permutation symmetries, and there is no solution to start from.

Question 10. Answer **C**.

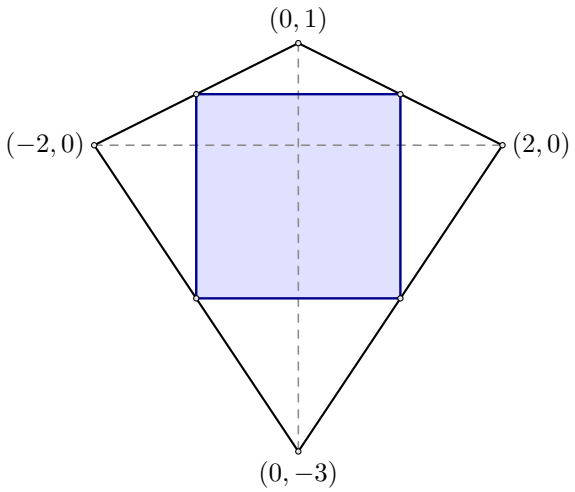


The figure shows both positions of D: the parallelogram on the left and the crossed quadrilateral on the right.

Triangles ABC and CDA share the side AC and have $AB = CD$, $BC = DA$, so they are congruent by SSS: III must hold. But congruence leaves two positions for D relative to the fixed triangle ABC: the reflection of B through the midpoint of AC, which makes $ABCD$ a parallelogram, and the reflection of B in the perpendicular bisector of AC, which puts D on the same side of AC as B. In the second case $ABCD$ is a crossed quadrilateral (its "sides" BC and DA cross), BD is parallel to AC, and neither I nor II holds. Only III must be true. The trap is that "two pairs of equal opposite sides" is a parallelogram criterion for convex quadrilaterals, and the question never said convex.

Question 11. Answer **A**.

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The midpoints of the sides of any quadrilateral form a parallelogram whose sides are parallel to the diagonals and half their length. That parallelogram is a square exactly when the diagonals are equal in length and perpendicular. A square has such diagonals, so the condition is necessary. It is not sufficient: the quadrilateral with vertices $(-2,0)$, $(0,1)$, $(2,0)$, $(0,-3)$ is convex, has perpendicular diagonals both of length 4, and is a kite, not a square. The condition is necessary but not sufficient.

Question 12. Answer **D**.

Each statement is a claim about remainders, and the recursion $a \mapsto a^2 - 2$ acts on remainders. Modulo 3: $4 \equiv 1$, then $1^2 - 2 \equiv 2$, then $2^2 - 2 = 2$, and the remainder stays 2 forever; no term is divisible by 3, so I holds. Modulo 12: $a_2 = 14 \equiv 2$, and $2^2 - 2 = 2$, so every later term is 2 (mod 12); II holds. Modulo 10: $4 \mapsto 16 - 2 = 14 \equiv 4$, so every term ends in 4; III holds. Modulo 5: the remainder is 4 forever, so no term is divisible by 5; IV fails. Three of the four are true. The first four terms $4, 14, 194, 37634$ confirm all of this, but the remainder argument is what makes it a proof rather than a pattern.

Question 13. Answer **E**.

I holds because $0 \le f \le 1$ gives $f^2 \le f$ pointwise, and integrating preserves the inequality. II is false: $f(x) = x$ gives $\int f^2 = \frac{1}{3}$ and $(\int f)^2 = \frac{1}{4}$. The true inequality runs the other way, and the neatest proof is worth knowing: with $m = \int_0^1 f$, expanding $0 \le \int_0^1 (f-m)^2 = \int f^2 - 2m^2 + m^2$ gives $\int f^2 \ge m^2$. For III, the hypothesis says $\int_0^1 (f - f^2) = 0$ with $f - f^2 = f(1-f) \ge 0$ continuous, so $f(1-f)$ vanishes everywhere: every value of f is 0 or 1. A continuous function taking only the values 0 and 1 cannot take both (it would have to pass through $\frac{1}{2}$), so f is constant. The answer is I and III. Option D is the reader who assumes an inequality between $\int f^2$ and $(\int f)^2$ without checking which way it goes.

Question 14. Answer **C**.

I: by the sine rule the sides are proportional to the sines of the opposite angles, so $\sin A + \sin B > \sin C$ is the triangle inequality $a + b > c$ in disguise. True. II: since $A + B < \pi$, we have $B < \pi - A$, and cosine is decreasing on $(0, \pi)$, so $\cos B > \cos(\pi - A) = -\cos A$. True. III: $\cos C = -\cos(A+B) = \sin A \sin B - \cos A \cos B$, so $\sin A \sin B > \cos C$ is equivalent to $\cos A \cos B > 0$, which fails whenever one of A, B is obtuse. For instance $A = 120°$, $B = C = 30°$ gives $\sin A \sin B = \frac{\sqrt{3}}{4} < \frac{\sqrt{3}}{2} = \cos C$. False. The answer is I and II. Each statement is a triangle fact wearing trigonometric clothing: the triangle inequality, the angle sum, and the sign of $\cos A \cos B$.

Question 15. Answer **E**.

I is impossible by counting black cells two ways: each row has at least 4 black cells, giving at least 24 in total, while each column has at most 2, giving at most 12. II is possible: colour the six cells

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on a diagonal black, so every row and column has exactly one. III is possible: colour black exactly the cells whose row number and column number are both even; any two consecutive rows contain exactly one even row and any two consecutive columns exactly one even column, so every 2×2 block contains exactly one black cell. The answer is II and III. Note that for II the parity check passes (six odd rows and six odd columns both give an even total) but passing a parity check is not a construction; the diagonal is.

Question 16. Answer **C**.

The claim is true, but the argument does not reach it. What has been shown is that a and b cannot both be non-multiples of 3; that leaves open the case where exactly one of them is, and the argument never mentions it. That case is easy to close (the remainders are then $0 + 1 = 1$, again not divisible by 3), but it has to be closed, and a proof that omits a case proves only the cases it treats. Options A and E are not counterexamples: $1 + 4 = 5$ is not divisible by 3, and $a = b = 3$ satisfies the conclusion. Option D is false since squares are 0 or 1 modulo 3. The point of the question is the difference between "at least one" and "both", which the student's final sentence quietly glosses over.

Question 17. Answer **A**.

Suppose S is non-empty. Fix any point O and, since S is finite, choose $P \in S$ furthest from O. If P were the midpoint of XY with $X, Y \in S$ distinct, then $\overrightarrow{OP} = \frac{1}{2}\big(\overrightarrow{OX} + \overrightarrow{OY}\big)$, and by the triangle inequality

$$OP \leq \tfrac{1}{2}(OX + OY) \leq \max(OX, OY),$$

with equality throughout only when $\overrightarrow{OX} = \overrightarrow{OY}$, i.e. $X = Y$. Since $X \neq Y$, one of OX, OY is larger than OP, contradicting the choice of P. So no non-empty finite set has the property, and S must be empty; the condition holds for the empty set vacuously. Options D and E fail for the same reason as everything else: in any finite set, the point furthest out is never a midpoint. Choosing an extreme element and showing it cannot satisfy the condition is the entire method.

Question 18. Answer **C**.

I: if f were never zero on $(0, 1)$ it would keep one sign there by continuity, and its integral could not be 0. So f changes sign, and has a zero. II: suppose f changes sign at only one point c, so f has one sign on $(0, c)$ and the opposite sign on $(c, 1)$. Then $(x - c)f(x)$ has the same sign on both intervals and is not identically zero, so $\int_0^1 (x - c)f(x)\,dx \neq 0$. But that integral equals $\int_0^1 xf - c\int_0^1 f = 0 - 0$, a contradiction. So f changes sign at least twice, giving at least two zeros; and if $f \equiv 0$ it has infinitely many. III fails: $f(x) = x^2 - x + \frac{1}{6}$ has $\int_0^1 f = \frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0$ and $\int_0^1 xf = \frac{1}{4} - \frac{1}{3} + \frac{1}{12} = 0$ without vanishing. The answer is I and II. The multiplier $(x - c)$ is the idea: it is a function of one sign built to be orthogonal to nothing but f.

Question 19. Answer **E**.

Write $3n + 1 = a^2$ and $4n + 1 = b^2$. Since $4n + 1$ is odd, b is odd, so $b^2 \equiv 1 \pmod 8$ and $4n \equiv 0 \pmod 8$: n is even. Then $3n + 1$ is odd, so a is odd and $a^2 \equiv 1 \pmod 8$, giving $3n \equiv 0 \pmod 8$, so $8 \mid n$. Now modulo 7: squares are $0, 1, 2, 4$, and $a^2 + b^2 = 7n + 2 \equiv 2$. The only way two squares sum to 2 modulo 7 is $1 + 1$, so $a^2 \equiv 1$, hence $3n \equiv 0 \pmod 7$ and $7 \mid n$. Therefore $56 \mid n$. And $n = 56$ itself works: $169 = 13^2$ and $225 = 15^2$, so no larger divisor is forced. The answer is 56. Option D is what one gets from the mod 7 argument together with only a partial mod 8 argument.

Question 20. Answer **A**.

Classify pile sizes as losing (L, the player to move loses) or winning (W). 0 and 1 are L, since no move is possible. Then $2 \to 0$ and $3 \to 1$ are W; $4 \to 1$ is W; 5 can only reach 3 and 2, both W, so 5 is L; 6 reaches 4 and 3, both W, so L; $7 \to 5$ is W; $8 \to 6$ is W; $9 \to 6$ is W; 10 reaches 8 and 7, both W, so L; 11 reaches 9 and 8, so L. The pattern is periodic with period 5: the losing sizes are those leaving remainder 0 or 1 on division by 5. Since $2026 = 5 \cdot 405 + 1$, the starting position is losing for the player to move: the second player wins. Options B to D assume the first player has a good move; option E misunderstands what best play means.

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